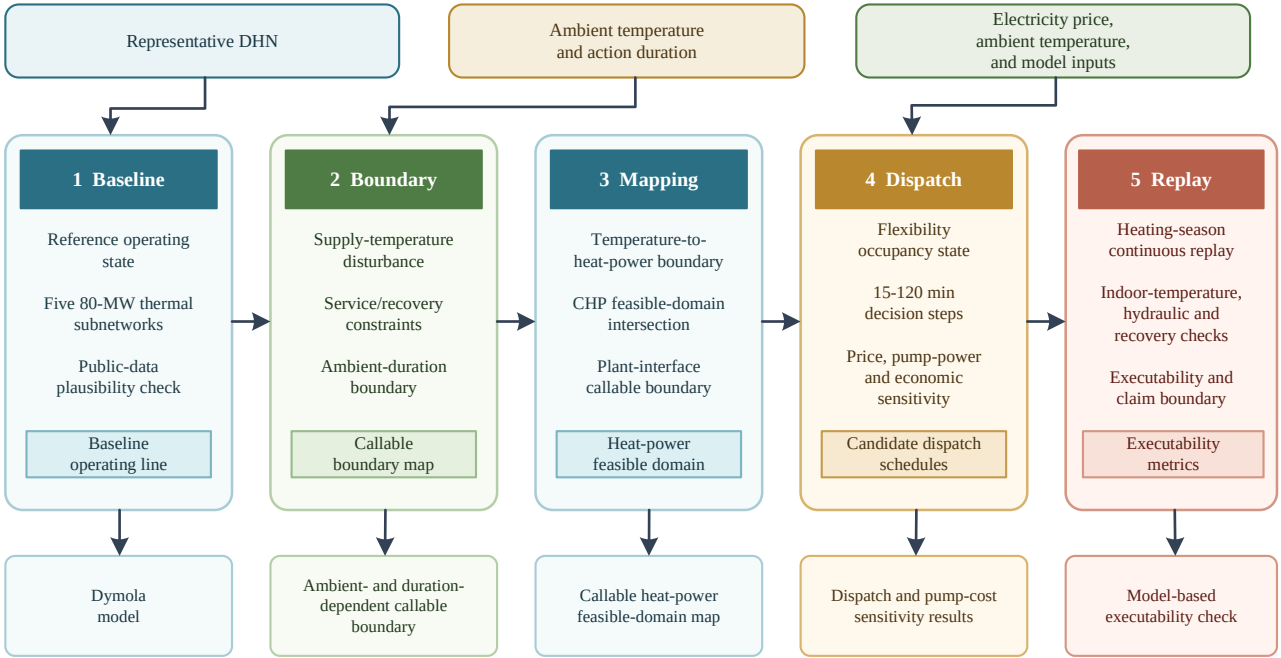
Representative DHN
Ambient temperature and action duration
Electricity price, ambient temperature, and model inputs
1 Baseline
Reference operating state
Five 80-MW thermal subnetworks
Public-data plausibility check
Baseline operating line
2 Boundary
Supply-temperature disturbance
Service/recovery constraints
Ambient-duration boundary
Callable boundary map
3 Mapping
Temperature-to-heat-power boundary
CHP feasible-domain intersection
Plant-interface callable boundary
Heat-power feasible domain
4 Dispatch
Flexibility occupancy state
15-120 min decision steps
Price, pump-power and economic sensitivity
Candidate dispatch schedules
5 Replay
Heating-season continuous replay
Indoor-temperature, hydraulic and recovery checks
Executability and claim boundary
Executability metrics
Dymola model
Ambient- and duration-dependent callable boundary
Callable heat-power feasible-domain map
Dispatch and pump-cost sensitivity results
Model-based executability check